Karamba3D

Twist Behaviour in Hybrid Truss

1 Document Information

Project Name	Hybrid Truss Example
Model / Example	ENG03 – Hybrid Truss
Software Versions	Karamba3D 3.1.60519.0
Author	Dr Dragos Naicu
Date	14.08.2026
Document Version	v01

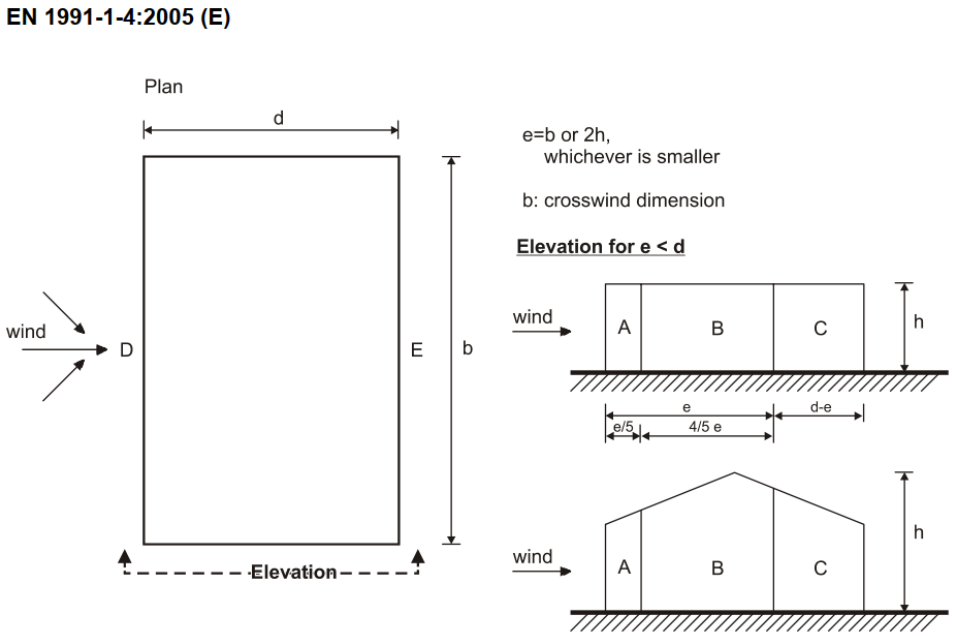
2 Model Overview

Structural System	Hybrid Glulam-Steel-Cable Truss
Geometry	Rectangular plan and arched profile
Size	44.5m main span with 6.0m braced bays
Material	Glulam, Steel
Cross-Sections	Rectangular, HEA, HEB, CHS, Round
Lateral System X	Braced frame
Lateral System Y	Portal frame truss
Number of Members	830
Number of Nodes	586
Software	Karamba3D 3.1. 60519.0 | Rhino 8 SR33 8.33.26188.13001
Model Source	Model inspired by StructureCraft project. https://structurecraft.com/projects/upper-macungie-township-community-recreation-center
Model Purposes	1. Explore „abnormal" behaviour in twisting of top chord of end bay trusses. 2. Validate model results with polyarc subdivision versus fine subdivison of top chord.

The example model was produced with several different purposes:

1) Build a complex hybrid material engineering example
2) Replicate a beautiful and elegant structural system based on a real case study
3) Build flat roof wind loading onto building according to Eurocode
4) Prove replicability of rectangular walls wind loading developed for example ENG02 Steel Tower

According to Eurocode, wind loading on a rectangular building is applied in zones A, B and C in areas perpendicular to the wind direction (suction on side walls) and zones D and E in areas along the wind direction (front wall pressure and back wall suction).



Eurocode 1991-1-4: wind zones on walls

Furthermore, for flat roofs, zones F, G, H and I are defined for application of pressure and suction, depending on the wind direction.

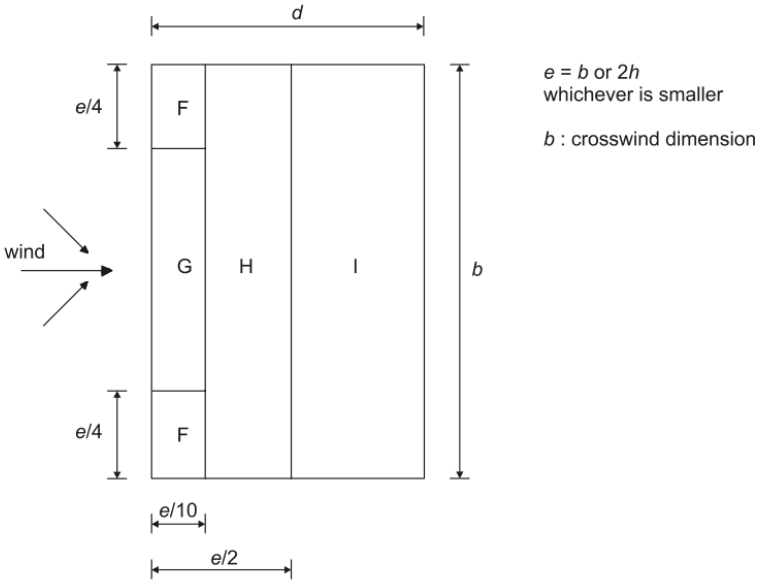
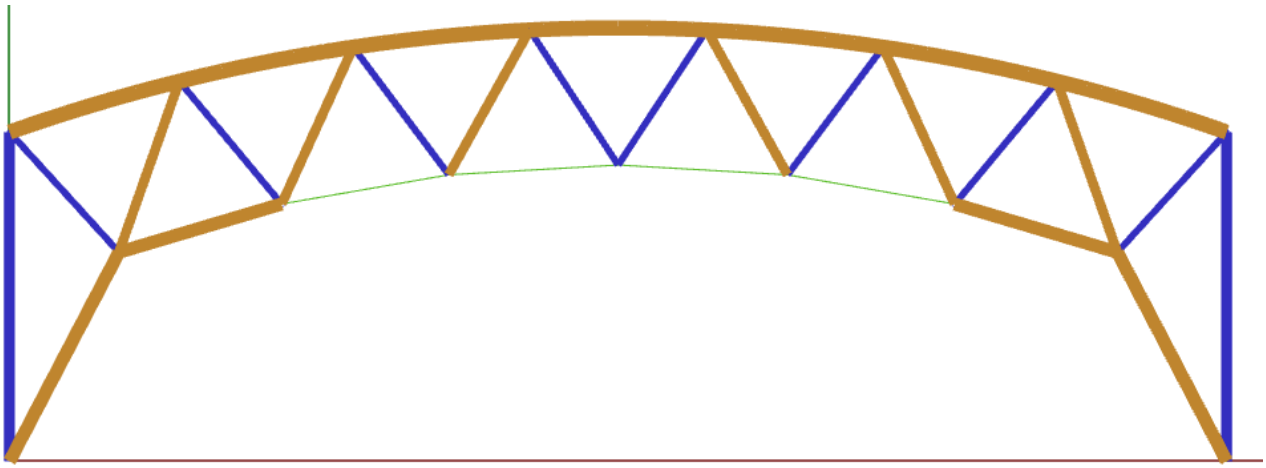


Figure 7.6 — Key for flat roofs

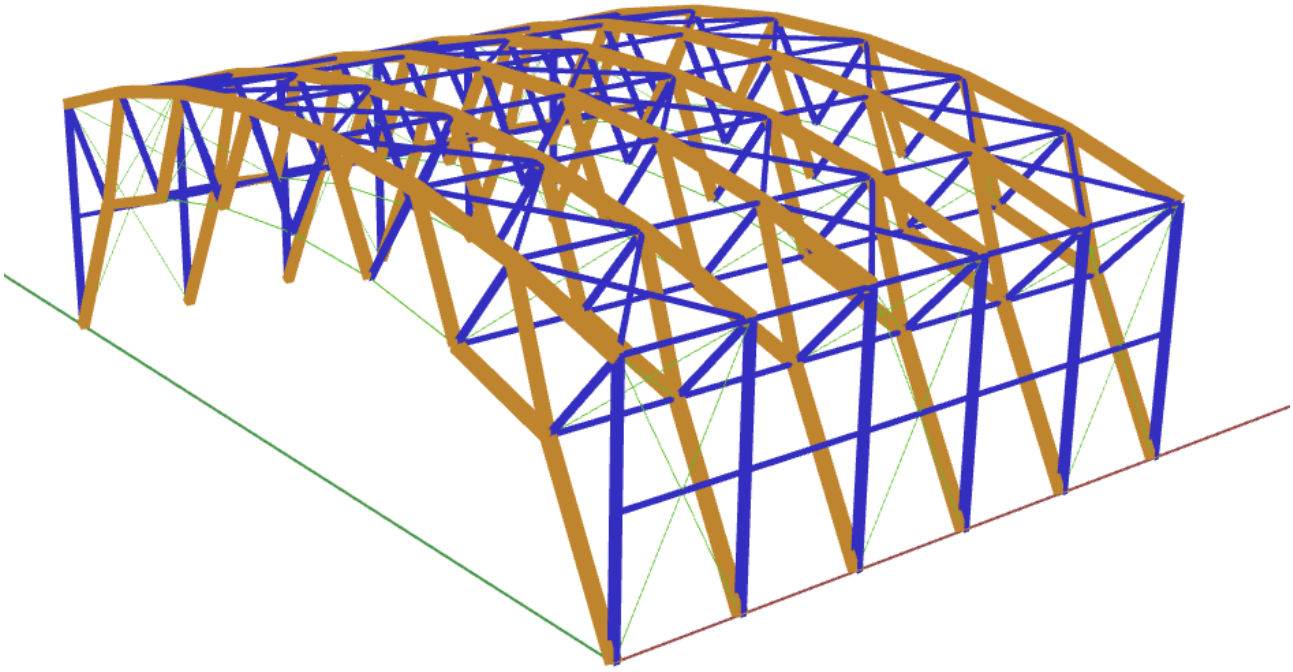
Eurocode 1991-1-4: wind zones on flat roof

The model is shown in the pictures below. The key feature is using a curved glulam top chord with a mix of steel and glulam diagonals and cables with prestress for the bottom chord. Additionally, the corners are framed with a mix of glulam and steel members, producing a portal frame truss system in Y.



Hybrid Truss view

To move into 3D, the example assumes a fixed bay width and lateral bracing every other bay. This uses diagonal CHS members on the roof and diagonal prestressed cables in the walls.



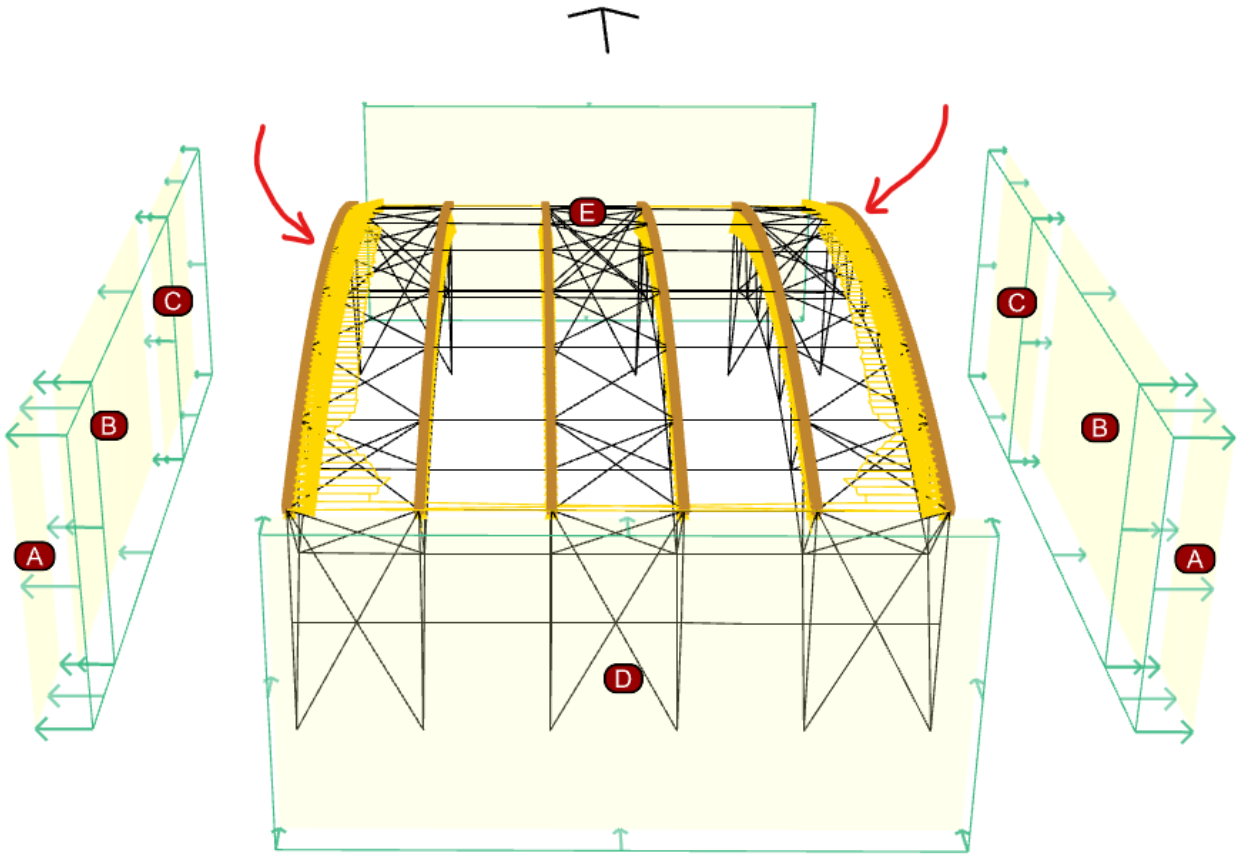
Overall System view

All dimensions are parametrically set up including span, number of bays, truss divisions and truss height at support and at midspan. Additionally, wind loads are fully parametrically defined.

3 Twist Behaviour in Top Chord

3.1 Loading Defintion

Because of the wind load definition, when wind is parallel to the trusses, there is suction (corresponding to zones A, B and C) applied laterally to the first and last curved glulam top chord. Image below shows line loads generated from mesh loads applied onto the top chords only. Notice decreasing magnitude from zone A to B to C.



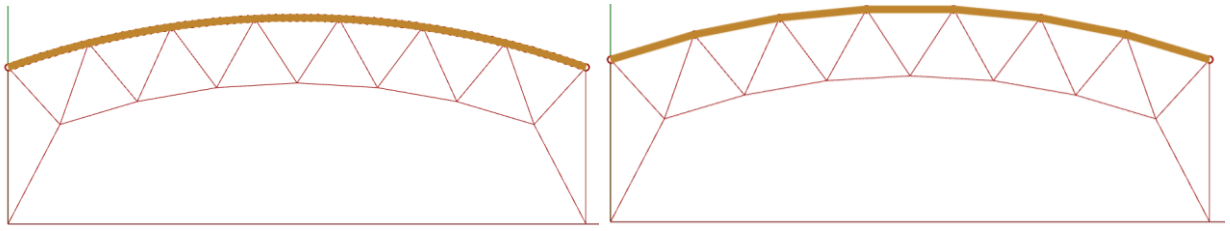
Loading of top chord due to wind laterally

3.2 Observation of „abnormal" twisting behaviour

The truss is divided based on the number of divisions selected. Leaving specific points where transverse connections and bracing are applied.

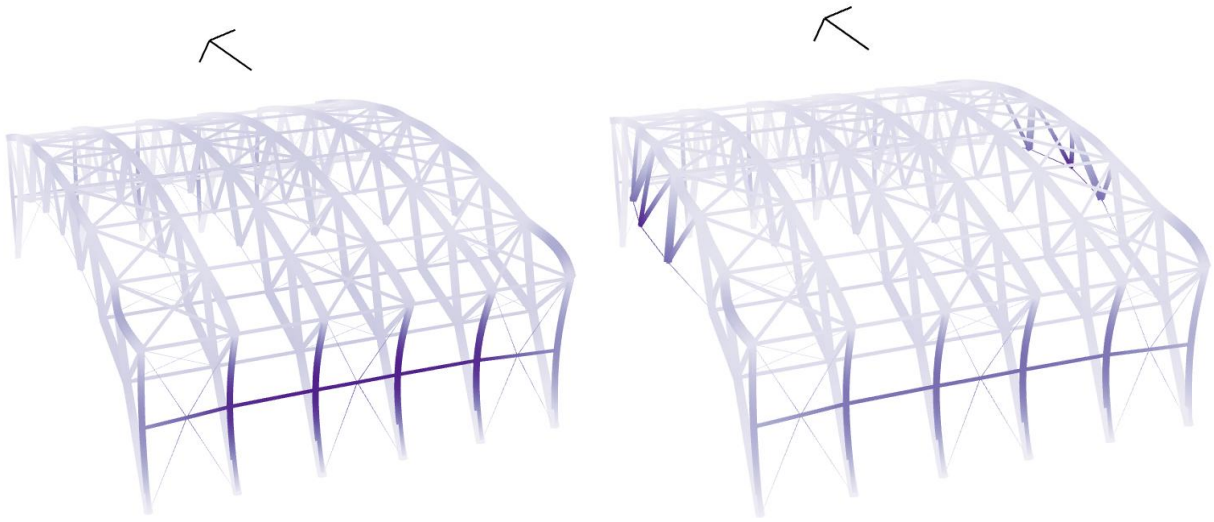
Additionally, one of the parameters available is the length divisions along the top chord. This subdivides the top chord sections into straight line elements for FE analysis.

In the extreme, you have 1 subdivision, meaning the top arc is approximated only by the truss divisions themselves. On the other extreme, you can have something like 0.5m producing a very smooth polyarc between each division point.



Comparison: Left showing smooth subdivided arc, Right showing polyline between truss divisions.

When exploring the parameter space, it was observed that in the case of a coarse polyline subdivision, there was a large twist of the truss happening. This was pointing outward on both sides. When the chord was more finely subdivided then there was no twist present and only lateral deformation as expected from the lateral line loads.



Comparison: Left showing smooth subdivided arc, Right showing polyline between truss divisions.

Note: this wind case contains also direct pressure on front wall and pressure/suction areas on the roof leading to complex deformation patterns in the overall system. However, the difference between the 2 situations, with one parameter change only is significant enough to invite further exploration of this behaviour.

4 Behaviour Investigation

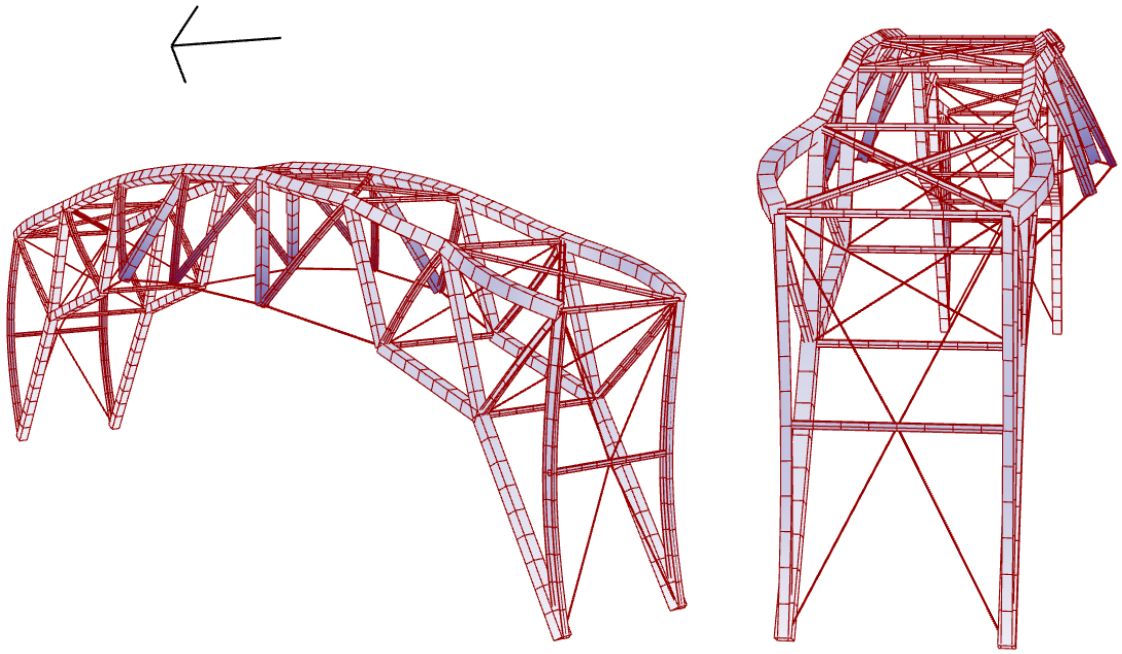
First, the loading conditions were investigated together with influence of prestress and gravity loads, etc.

Secondly, a review of releases in the structure was performed with no obvious cause identified.

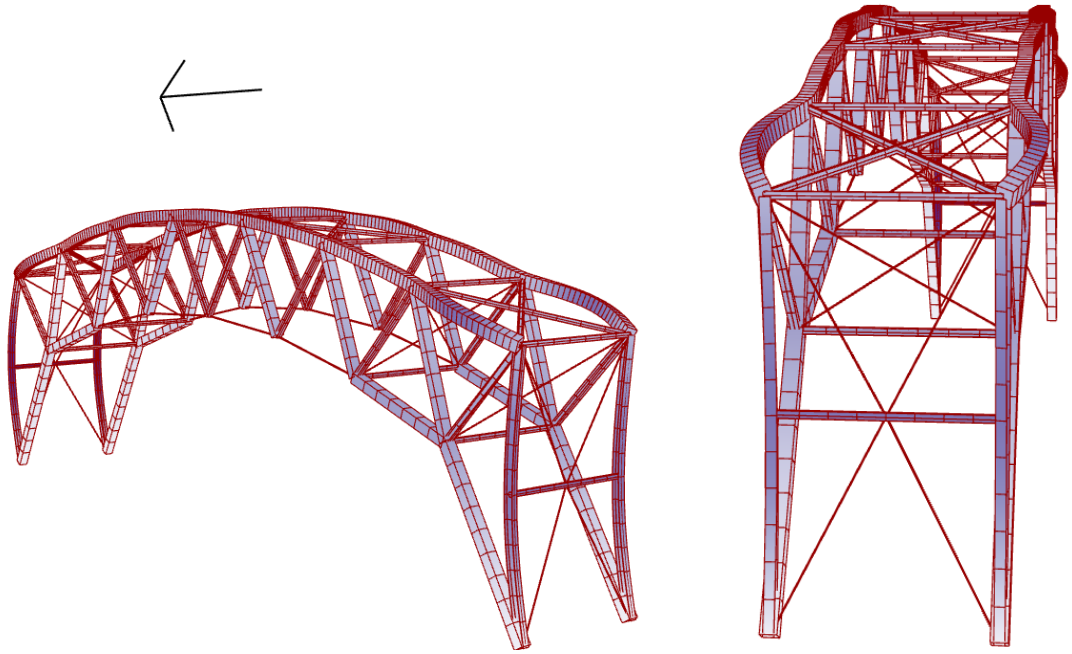
Then reductions and simplifications are applied as described below.

4.1 Step 1 – Look at 1 bay only

In the first step, the structure was reduced to a single bay, containing 2 truss frames.
The behaviour was clearly observed, as shown below. The polyline version has both trusses flaring outwards whereas the subdivided version behaves very differently. There is significant out of plane deformation, particularly in the first division which would be loaded according to zone A, the highest value of suction. The same lateral deformation is present with the polyline version, but this one also presents the large twist.



Polyline version: shows top chord twist outward and induced rotations in the truss leading to the bottom chord moving outward.

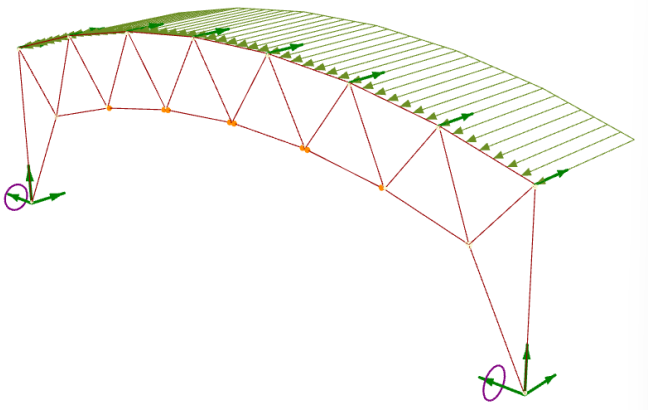


Subdivided version: shows no top chord twist and all deformations corresponding to lateral suction forces pulling on the top chord.

This started to indicate potential causes related to the truss itself. See next step.

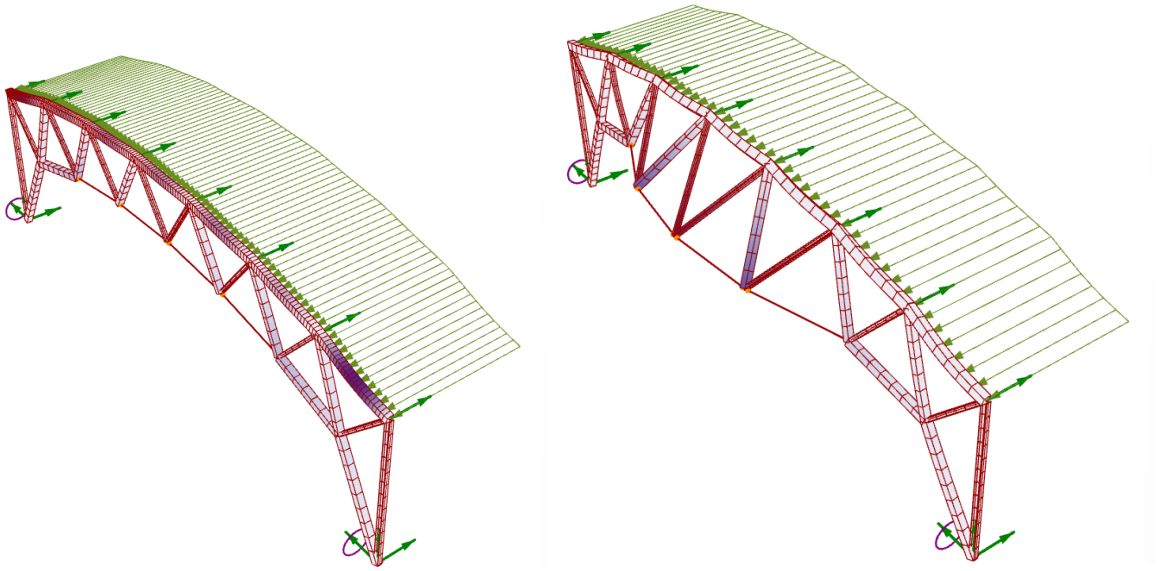
4.2 Step 2 – Look at 1 truss only

The next step was to isolate a single truss. The loading was simplified to have only lateral uniform line load on the top chord. To stabilise the system in 3D, rotation out of plane was restrained at the supports and lateral supports were added on the top chord at the division points. This mirrors the location of transverse and bracing elements in the 3D structure.

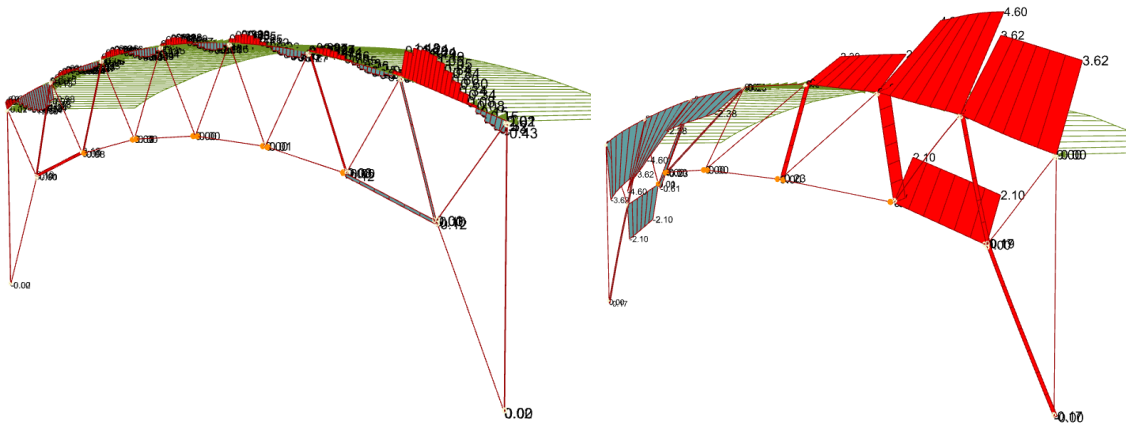


Simple 2D truss system

The behaviour was again observed as shown below.



Comparsion: Left – subdivided version with no twist; Right – polyline version with twist

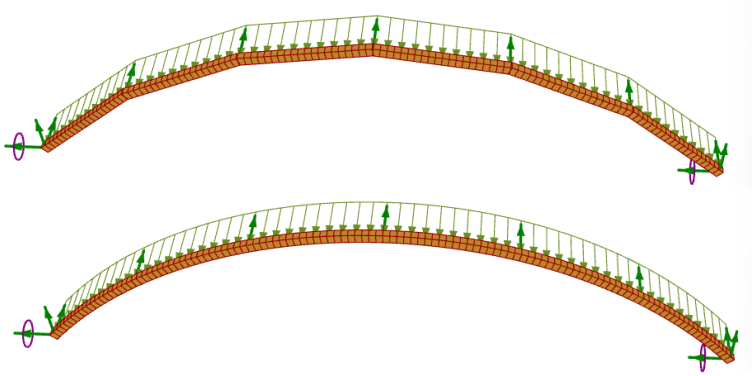


Comparsion: Left – subdivided version with no twist; Right – polyline version with twist
Left – shows smaller and smoother Mx results; Right – shows high Mx results, uniform per chord segment

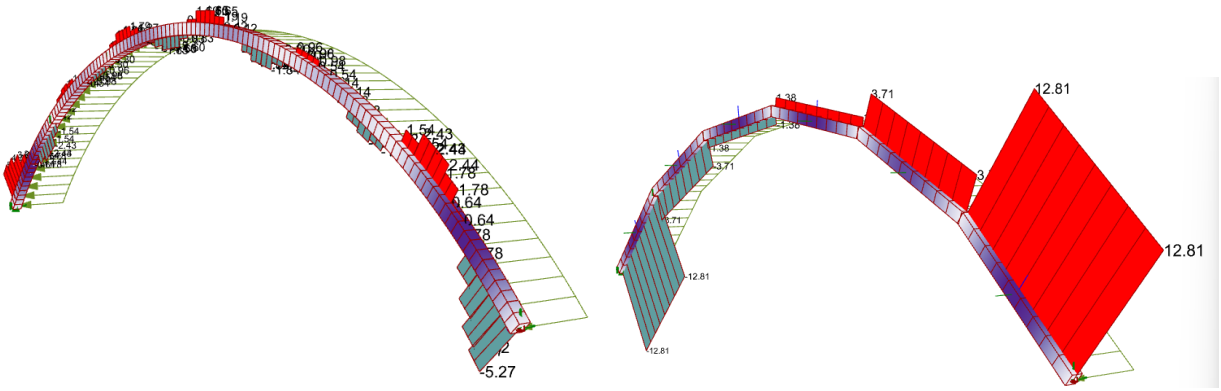
At this point, it was apparent that the subdivison level in a curved planar beam coupled with lateral loading perpendicular to the beam plane is an actual factor in determining the deformation and additional twist or torsion present in the beam. The continuity across sharp kinks and loading perpendicular to that kink are connected. The next simplification is to just isolate the top chord, as explained below.

4.3 Step 3 – Look at top chord only

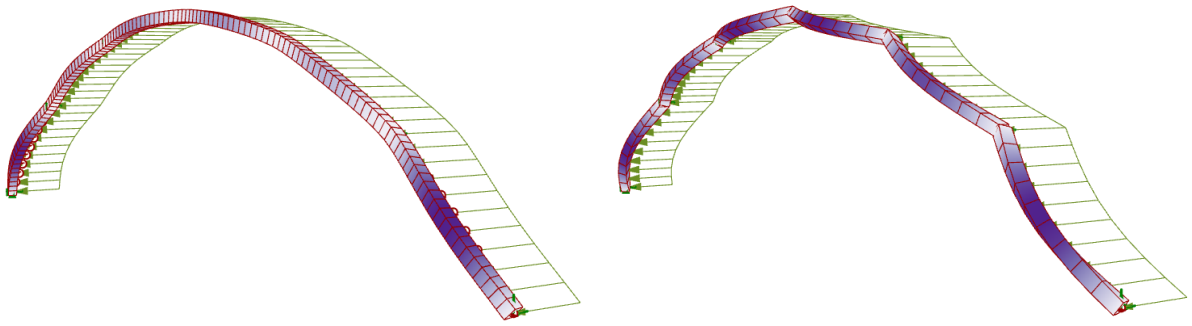
The final experiment is just a simple arc that is parametrised based on span and height. Additionally we have number of subdivisions, with lateral restraints at the subdivision points. And the experimental choices are between polyline and subdivided arc, and number of subdivisions/segment.



Simple 2D arc system (polyline top and smooth subdivided arc bottom)

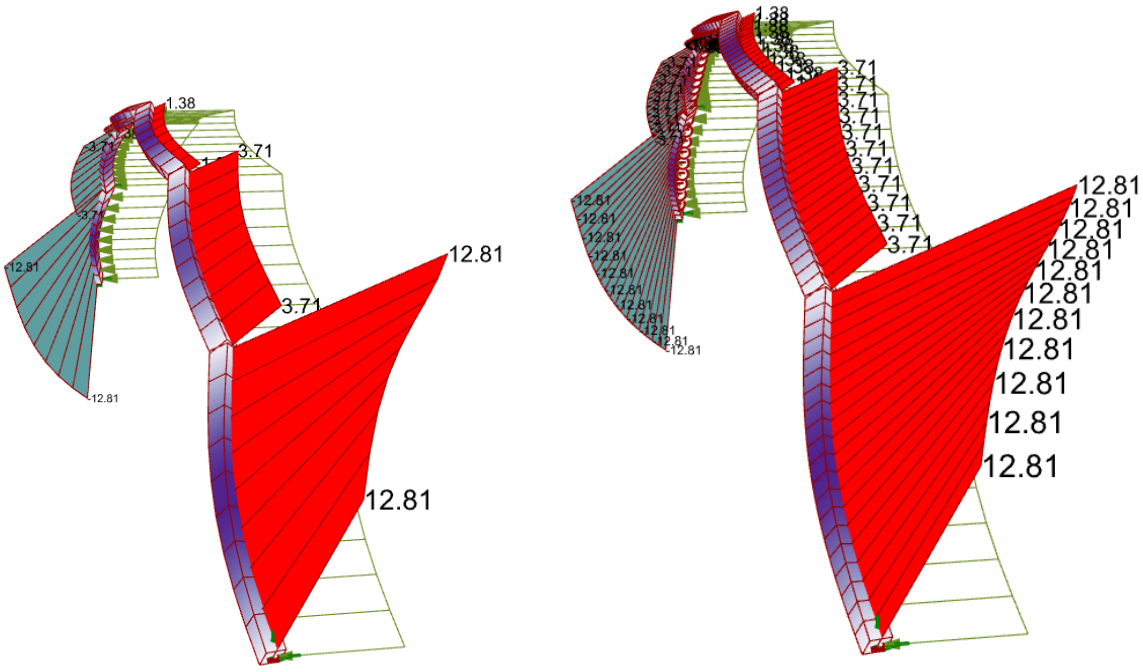


Comparsion: Left – subdivided version; Right – polyline version (showing Mx on undeformed structure)
Left – shows smaller and smoother Mx results; Right – shows high Mx results, uniform per chord segment



Comparsion: Left – subdivided version; Right – polyline version (deformation 50x scale)
Left – shows primarily out of plane bending; Right – shows out of plane bending + twist

To further investigate, the polyline version was also subdivided into higher and higher segment numbers (up to 20) for each line, which produces identical results to the non-subdivided polyline (i.e. large twist and Mx).

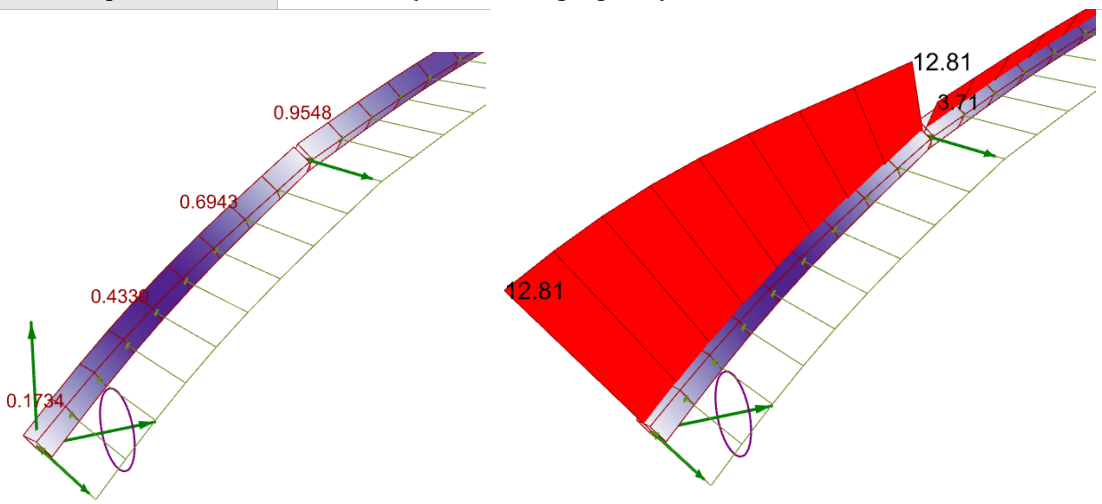

1.38
3.71
12.81
3.71
12.81
12.81
12.81

5 Additional Verification - Twist

Karamba3D's Beam Displacements component reports nodal rotations in global X/Y/Z coordinates, not in the beam's local frame. To recover twist about the beam axis (local x), the global rotation vector is dotted with the beam's local-x unit vector. We now verify that extracted twist against a closed-form Saint-Venant hand calculation for a single straight, prismatic segment under constant torque.

5.1 Inputs

Section	Rectangular glulam, GL28h — 56 cm × 48 cm
Shear modulus, G (Karamba3D)	78 kN/cm^2
Torsional constant, J (Karamba3D)	1.0053 × 10^6 cm^4
Segment length, L	8.3506 m (straight)
Internal torque, Mx	12.81 kNm (constant along segment)



Left: Rotation rx values along first segment in local frame. Right: Constant Mx value for first segment.

5.2 Torsional Constant J

Independent check using Saint-Venant's approximate formula for a solid rectangle (a = long side, b = short side):

$\beta = 1/3 - 0.21 \cdot (b/a) \cdot [1 - b^4/(12a^4)] = 0.1614$

$J = \beta \cdot a \cdot b^3 = 0.1614 \times 56 \times 48^3 \approx 999{,}600 \text{ cm}^4$

Matches K3D reported 1.0053 × 10^6 cm^4 within ~0.9%. Warping constant = 0 is as expected.

5.3 Twist Calculation

Saint-Venant torsion, no distributed torque along the segment — twist rate is constant:

$GJ = G \cdot J = 78 \times 1.0053 \times 10^6 = 78{,}413{,}400 \text{ kN} \cdot \text{cm}^2 = 7{,}841.34 \text{ kN} \cdot \text{m}^2$

$d\theta/dx = Mx / GJ = 12.81 / 7{,}841.34 = 1.6336 \times 10^{-3} \text{ rad/m} \quad (0.0936 °/\text{m})$

$\theta = d\theta/dx \cdot L = (0.0936 °/\text{m} \times 8.3506 \text{ m}) = 0.7817°$

5.4 Comparison to Model

From the model we extract the total twist based on the end – start rotation angles. These are converted to the local frame by dotting the global rotation with the beam's local-x unit vector. Values are shown below.

Source	Twist across segment
Hand calc (Saint-Venant)	0.7817°
Model (global → local-x dot product)	0.9548° – 0.1734° = 0.7814°
Difference	**0.0003° (~0.03%)**

5.5 Conclusion

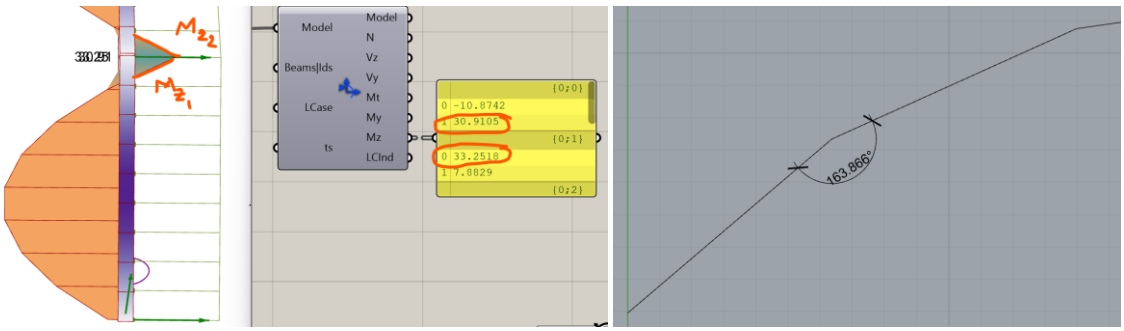
Overall Validation Result	PASS
Comments / Remarks	The reported twist values resulting from the torsional moment are confirmed against simple hand calculation based on section properties and moment. This indicates that the top chord behaviour is actually correct when isolated down to the simple polyline arc under lateral loading. In the truss context, the top chord twist then rotates the diagonals and bottom chord of the system outward and the full 3D behaviour is correct.
Author	Dr Dragos Naicu Date: 14.08.2026

6 Additional Verification – Moment across kink

The final bit of verification involves attempting to derive the actual Mx values based on the lateral load. We confirm that under the constant lateral load, the change in orientation at the kink is what creates the high and realistic Mx values in the polyline arc, which in turn lead to the overal twist of the top chord in the truss. M (the physical internal moment vector) is continuous through the kink node; only the local (x, z) frame rotates by Δφ, converting bending (Mz) into torque (Mx) via the sin(Δφ) term.

6.1 Inputs

Kink angle	Δφ = 180° – α = 180° – 163.866° = 16.134°
Out-of-plane moment Mz before kink [kNm]	30.9105 kNm
Out-of-plane moment Mz after kink [kNm]	33.2518 kNm



Left: Mx values from model. Right: Kink angle value from model

$\Delta\phi = 180° - \alpha = 180° - 163.866° = 16.134°$
$\sin(\Delta\phi) = 0.27788$
$\cos(\Delta\phi) = 0.96060$

$Mx_1 = [Mz_1 \cdot \cos(\Delta\phi) - Mz_2] / \sin(\Delta\phi)$
$= [30.9105 \times 0.96060 - 33.2518] / 0.27788$
$= -12.809$ kNm → Karamba: 12.81 kNm ✓

$Mx_2 = Mx_1 \cdot \cos(\Delta\phi) + Mz_1 \cdot \sin(\Delta\phi)$
$= (-12.809)(0.96060) + (30.9105)(0.27788)$
$= -3.713$ kNm → Karamba: 3.71 kNm ✓

Check: $|M| = \sqrt{(Mx^2 + Mz^2)}$
before: $\sqrt{(12.809^2 + 30.9105^2)} = 33.46$
after: $\sqrt{(3.713^2 + 33.2518^2)} = 33.46$ (magnitude conserved ✓)

6.2 Comparison to Model

Using the Karamba3D Mz values before and after the kink, and the kink angle, we determine what Mx should be in the 2 segments.

From the model we extract Mx before the kink and Mx after kink (constant along each segment of the polyline).

Source	Mx moments across kink
Model Mx1	12.81 kNm
Hand Calc Mx1	12.809 kNm
Model Mx2	3.71 kNm
Hand Calc Mx2	3.713 kNm

6.3 Conclusion

Overall Validation Result	PASS
Comments / Remarks	The generated Mx values are consistent with the geometry and loading and prove that lateral loading combined with orientation changes at kinks converts some Mz bending into torsional Mx. This in turn produces correct twisting values which explain the initially observed "abnormal" behaviour. When laterally loading a polyline continuous beam, development of actual twisting and torsion because of the severity of the kinks is something an engineer must account for. When approximating a curved continuous beam, it should be done such that the additional twisting behaviour that would not be real for the real curved beam does not show up in the analysis and induce additional design penalties.
Author	Dr Dragos Naicu Date: 14.08.2026